

Practical No. : 4

Aim :- Analyze the packets provided in lab and solve the questions using Wireshark :

1. Recover the username and password from the insecure website “zerobank” using Wireshark

Analysis –

Not secure

zero.webappsecurity.com/login.html?login_error=true

Zero Bank

Troubles entering the site?

Login and/or password are wrong.

Login

Password

Keep me signed in

☐

Sign in

[Forgot your password ?](#)

No.	Time	Source	Destination	Protocol	Length	Text item	Host	Protocol	Request Method	Info
12933	65.356063	54.82.22.214	172.30.1.95	TCP	1197	✓				80 → 62214 [PSH, ACK] Seq=17355 Ack=611 Win=64128 Len=1
13044	65.553512	54.82.22.214	172.30.1.95	TCP	1514	✓				80 → 62214 [ACK] Seq=10678 Ack=611 Win=64128 Len=1
13045	65.553621	54.82.22.214	172.30.1.95	TCP	1514	✓				80 → 62214 [ACK] Seq=20138 Ack=611 Win=64128 Len=1
13573	70.166766	172.30.1.95	54.82.22.214	HTTP	573	✓	zero.webappsecu...	GET		GET /login.html HTTP/1.1
13611	70.440312	54.82.22.214	172.30.1.95	TCP	1514	✓				80 → 62235 [ACK] Seq=102223 Ack=961 Win=64128 Len=1
13613	70.440415	54.82.22.214	172.30.1.95	TCP	1514	✓				80 → 62235 [ACK] Seq=103683 Ack=961 Win=64128 Len=1
13627	70.565249	172.30.1.95	54.82.22.214	HTTP	369	✓	zero.webappsecu...	GET		GET /login.html HTTP/1.1
13644	70.824681	54.82.22.214	172.30.1.95	TCP	1514	✓				80 → 62214 [ACK] Seq=25985 Ack=926 Win=64128 Len=1
13645	70.824681	54.82.22.214	172.30.1.95	TCP	1514	✓				80 → 62214 [ACK] Seq=27445 Ack=926 Win=64128 Len=1
13871	74.140505	122.15.192.126	172.30.1.95	TCP	1514	✓				80 → 62227 [ACK] Seq=9991 Ack=313 Wins=64128 Len=146
13876	74.144616	172.30.1.95	122.15.192.126	HTTP	366	✓	agriclicense.upag...	GET		GET /live/ZB-zerobank-in-India.html HTTP/1.1
14238	80.784742	172.30.1.95	54.82.22.214	HTTP	817	✓	zero.webappsecu...	POST		POST /signin.html HTTP/1.1 (application/x-www-form-urlencoded)
14800	89.500760	122.15.192.126	172.30.1.95	TCP	1514	✓				80 → 62227 [ACK] Seq=21557 Ack=625 Win=64128 Len=1
18889	140.346664	172.30.1.95	54.82.22.214	HTTP	626	✓	zero.webappsecu...	GET		GET /login.html?login_error=true HTTP/1.1
18925	140.607229	54.82.22.214	172.30.1.95	TCP	1514	✓				80 → 62235 [ACK] Seq=110255 Ack=2296 Win=64128 Len=1
19009	141.039982	172.30.1.95	54.82.22.214	HTTP	386	✓	zero.webappsecu...	GET		GET /login.html?login_error=true HTTP/1.1
19062	141.551775	54.82.22.214	172.30.1.95	TCP	1514	✓				80 → 62255 [ACK] Seq=325 Ack=333 Win=64128 Len=146

> Frame 14238: 817 bytes on wire (6536 bits), 817 bytes captured (6536 bits) on interface \Device\NPF_{DA...}

> Ethernet II, Src: Dell 31:08:ef (30:0d:42:31:08:ef), Dst: AcrosserTech_46:43:a2 (00:02:b6:46:43:a2)

> Internet Protocol Version 4, Src: 172.30.1.95, Dst: 54.82.22.214

> Transmission Control Protocol, Src Port: 62235, Dst Port: 80, Seq: 961, Ack: 109561, Len: 763

> Hypertext Transfer Protocol

> HTML Form URL Encoded: application/x-www-form-urlencoded

```

0000  00 02 b6 46 43 a2 30 d0 42 31 08 ef 08 00 45 00  --FC-0: B1-----E-
0010  03 23 cb 28 40 00 00 06 00 00 ac 1e 01 5f 36 52  -#(@-.....6R
0020  16 06 f3 1b 00 50 7f 76 2d 62 30 8b 5c 0c 50 18  -.....P.....
0030  02 00 fd ba 00 00 50 4f 53 54 20 2f 73 69 6e 00  -.....PO ST /sign
0040  69 6e 20 68 74 6d 6c 20 48 54 54 50 2f 31 2e 31  -in.html HTTP/1.1
0050  0d 04 68 f7 73 3a 20 7a 65 72 6f 2e 77 65 62 02  -Host: zero.web
0060  61 70 7f 63 65 63 75 72 69 74 79 2e 63 6f 6d 0d  -appsecu rity.com
0070  04 73 6f 6e 6e 65 63 7a 69 6f 6e 3a 20 6b 65 65  -Connect ion: kee
0080  7d 20 61 6c 69 76 65 0d 04 43 6f 6e 74 65 6e 74  -p-alive- Content
0090  2d 4c 65 6e 67 74 68 3a 20 39 37 0d 0a 43 61 6f  -Length: 97- Cac
00a0  68 65 2d 43 6f 6e 74 6f 6e 6c 3a 20 6d 61 78 2d  -he(Contr ol: max-
00b0  61 67 65 3d 30 0d 04 4f 72 69 6f 69 6e 3a 20 68  -age=0-Orig in: h
00c0  74 74 70 3a 2f 2f 7a 65 72 6f 2e 77 65 62 61 70  -ttp://ze ro.webap
00d0  70 73 65 63 75 72 69 74 79 2e 63 6f 6d 0d 0a 43  -psecu rity.com-C
00e0  6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20 61 70 70  -ontent-1 ype: app
00f0  6c 69 63 61 74 69 6f 6e 2f 7d 72 77 77 2d 66 6f  -lication /x-ww-fo
0100  6f 72 6d 2d 75 72 6e 65 6e 63 6f 6e 64 65 0d 04  -orm-url-ncode d-
0110  55 70 67 72 61 64 65 2d 49 6e 73 65 63 75 72 65  -Upgrade- Insecu
0120  2d 52 65 61 75 65 73 74 73 3a 20 31 0d 0a 55 73  -Request- s: 1-Us
0130  65 72 2d 41 71 67 65 6e 74 3a 20 4d 6f 7a 69 6c  -er-Agent : Mozill

```

Search for “post request in http in ZeroBank”
We can see the username and password in HTTP

Capturing on Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Frame matches "zero"

No.	Time	Source	Destination	Protocol	Length	Text Item	Host	Hypertext Transfer Protocol	Request Method	Info
12933	65.356663	54.82.22.214	172.30.1.95	TCP	1197	✓				80 → 62214 [PSH, ACK] Seq=17535 Ack=611 Win=64128 Len=1
13844	65.553512	54.82.22.214	172.30.1.95	TCP	1514	✓				80 → 62214 [ACK] Seq=18678 Ack=611 Win=64128 Len=1
13845	65.553621	54.82.22.214	172.30.1.95	TCP	1514	✓				80 → 62214 [ACK] Seq=20138 Ack=611 Win=64128 Len=1
13573	70.166766	172.30.1.95	54.82.22.214	HTTP	573	✓	zero.webappsecu...		GET	GET /login.html HTTP/1.1
13611	70.440312	54.82.22.214	172.30.1.95	TCP	1514	✓				80 → 62235 [ACK] Seq=182223 Ack=961 Win=64128 Len=1
13613	70.440415	54.82.22.214	172.30.1.95	TCP	1514	✓				80 → 62235 [ACK] Seq=183683 Ack=961 Win=64128 Len=1
13627	70.565249	172.30.1.95	54.82.22.214	HTTP	369	✓	zero.webappsecu...		GET	GET /login.html HTTP/1.1
13644	70.824681	54.82.22.214	172.30.1.95	TCP	1514	✓				80 → 62214 [ACK] Seq=25985 Ack=926 Win=64128 Len=1
13645	70.824681	54.82.22.214	172.30.1.95	TCP	1514	✓				80 → 62214 [ACK] Seq=27445 Ack=926 Win=64128 Len=1
13871	74.140595	122.15.192.126	172.30.1.95	TCP	1514	✓				80 → 62227 [ACK] Seq=5991 Ack=313 Win=64128 Len=1
13876	74.144616	172.30.1.95	122.15.192.126	HTTP	366	✓	agriclicense.upag...		GET	GET /live/28-zero-bank-in-India.html HTTP/1.1
14238	80.784742	172.30.1.95	54.82.22.214	HTTP	817	✓	zero.webappsecu...		POST	POST /signin.html HTTP/1.1 (application/x-www-form-urlencoded)
14800	89.500760	122.15.192.126	172.30.1.95	TCP	1514	✓				80 → 62227 [ACK] Seq=21557 Ack=625 Win=64128 Len=1
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18925	140.607229	54.82.22.214	172.30.1.95	TCP	1514	✓				80 → 62235 [ACK] Seq=110255 Ack=2296 Win=64128 Len=1
19009	141.039982	172.30.1.95	54.82.22.214	HTTP	386	✓	zero.webappsecu...		GET	GET /login.html?login_error=true HTTP/1.1
19062	141.551775	54.82.22.214	172.30.1.95	TCP	1514	✓				80 → 62255 [ACK] Seq=325 Ack=333 Win=64128 Len=1

> Frame 14238: 817 bytes on wire (6536 bits), 817 bytes captured (6536 bits) on interface \Device\NPF_{D4000000-0000-0000-0000-000000000000} (00:02:b6:46:a3:a2) (00:02:b6:46:a3:a2)

> Ethernet II, Src: Dell_31:08:ef (30:0d:42:31:08:ef), Dst: AcrosserTech_46:a3:a2 (00:02:b6:46:a3:a2)

> Internet Protocol Version 4, Src: 172.30.1.95, Dst: 54.82.22.214

> Transmission Control Protocol, Src Port: 80, Dst Port: 80, Seq: 961, Ack: 109561, Len: 763

> Hypertext Transfer Protocol

> HTML Form URL Encoded: application/x-www-form-urlencoded

> Form item: "user_login" = "abcd"

> Form item: "user_password" = "1234"

> Form item: "submit" = "Sign in"

> Form item: "user_token" = "%29fa3b5e-e725-4cd5-92fd-aea348a5ee69"

0000 00 02 b6 46 a3 a2 30 d0 42 31 08 ef 00 00 45 00 ...FC 00 B1 ...E-

0010 03 23 cb 28 40 00 80 06 00 00 ac 1e 01 5f 36 52 ...-# (@... ..-G8

0020 16 d6 f3 1b 00 50 7f 76 2d 62 30 8b 5c 0c 50 18 ...-P v -b0 \P-

0030 02 00 fd ba 00 00 50 4f 53 54 20 2f 73 69 67 6e ...-PO ST /sign

0040 69 6e 2e 68 74 6d 6c 20 48 54 50 2f 31 2a 31 ...in.html HTTP/1.1

0050 0d 0a 4b 6f 73 74 3a 20 7a 65 72 6f 2e 77 65 62 ...-Host: zero.web

0060 61 70 70 73 65 63 75 72 69 74 79 2e 63 6f 6d 0d ...appsecu ity.com

0070 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20 6b 65 65 ...-Connect ion: kee

0080 70 2d 61 6c 69 76 65 6d 0a 43 6f 6e 74 65 6e 74 ...p-alive: Content

0090 2d 4c 65 6e 67 74 68 3a 20 35 37 0d 0a 43 61 63 ...Length: 97 -Cac

00a0 68 65 2d 43 6f 6e 74 72 6f 6e 3a 20 6d 61 78 2d ...he-Contr ol: max

00b0 61 67 65 3d 30 0d 0a 4f 72 69 67 69 6e 3a 20 68 ...age=0 0 rigin: h

00c0 74 74 70 3a 2f 2f 7a 65 72 6f 6e 77 65 62 61 70 ...ttp://ze ro.weba

00d0 70 73 65 63 75 72 69 74 79 2e 63 6f 6d 0d 0a 43 ...psecu it y.com -C

00e0 6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20 61 70 70 ...ontent-T ype: app

00f0 6c 69 63 61 74 69 6f 6e 2f 78 2d 77 77 77 2d 66 ...lication /x-www-f

0100 6f 72 6d 2d 75 72 6c 65 6a 63 6f 64 65 64 6d 0a ...om-urle ncoded:f

0110 55 70 67 72 61 64 65 2d 49 6e 73 65 63 75 72 65 ...Upgrade -Insecure

0120 2d 52 65 71 75 65 73 74 73 3a 20 31 0d 0a 55 73 ...-Request s: 1 -Us

0130 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6e ...er-Agent : Mozill

0140 61 2f 35 2e 30 20 28 57 69 6e 64 6f 77 73 20 4e ...a/5.0 (Windows N

0150 54 20 31 30 2e 30 3b 20 57 69 6e 36 3a 3b 20 78 ...T 10.0; Win64; x

0160 36 34 29 20 41 70 79 6c 65 57 65 62 4b 69 74 2f ...64) Appl ewebkit/

Packets: 36059 Displayed: 43 (0.1%)

2. How many web servers are running “gws”?

Analysis – The web server name can be retrieved from HTTP response header. So will apply filter http.response and we can see all http response packets.

*Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http.response.code==200

No.	Time	Source	Destination	Protocol	Length	Text Item	Host	Info
1646	1.320970	23.57.214.214	172.30.1.37	HTTP	1395	✓		HTTP/1.1 200
2156	1.833790	23.54.82.195	172.30.1.37	HTTP	1297	✓		HTTP/1.1 200

> Frame 1646: 1395 bytes on wire (11160 bits), 1395 bytes captured (11160 bits) on interface \Device\NPF_{D4000000-0000-0000-0000-000000000000} (00:02:b6:46:a3:a2) (00:02:b6:46:a3:a2)

> Ethernet II, Src: AcrosserTech_46:a3:a2 (00:02:b6:46:a3:a2), Dst: Dell_31:08:ef (30:0d:42:31:08:ef)

> Internet Protocol Version 4, Src: 23.57.214.214, Dst: 172.30.1.37

> Transmission Control Protocol, Src Port: 80, Dst Port: 80, Seq: 255, Ack: 255, Len: 0

> [2 Reassembled TCP Segments (4096 bytes total)]

> Hypertext Transfer Protocol

> HTTP/1.1 200 OK\r\n

[Expert Info (Chat/Sequence)]

[HTTP/1.1 200 OK\r\n]

[Severity level: Chat]

[Group: Sequence]

Response Version: HTTP/1.1

Status Code: 200

[Status Code Description: OK]

Response Phrase: OK

Server: nginx\r\n

Content-Type: application/pkix-cert\r\n

Last-Modified: Fri, 04 Aug 2023 20:57:56 GMT\r\n

Etag: "64cd6654-56f"\r\n

Content-Disposition: attachment; filename="ISRG Root X1.der"\r\n

0010 0a 53 65 72

0020 43 6f 6e 74

0030 70 6c 69 63

0040 65 72 74 0d

0050 65 64 3a 20

0060 32 30 32 33

0070 54 0d 0a 45

0080 35 34 2d 35

0090 2d 44 69 73

00a0 74 61 63 68

00b0 6d 65 3d 22

00c0 2e 64 65 72

00d0 6e 67 65 73

00e0 79 3a 20 41

00f0 6e 67 0d 0a

0100 64 69 6e 67

0110 65 6e 74 2d

0120 0d 0a 43 61

0130 20 6d 61 78

0140 45 78 70 69

0150 20 4a 61 6e

Now we will set the server header as column select any packet and right click on it then select Apply as Column.

Wireshark interface showing packet 113512 selected. The filter is 'http.server contains "gws"'. The packet details pane shows the following information:

- Frame 113512: 1111 bytes on wire (8888 bits), 1111 bytes captured (8888 bits) on
- Ethernet II, Src: AcrosserTech_46:43:a2 (00:02:b6:46:43:a2), Dst: LiteOnNetwor_6
- Internet Protocol Version 4, Src: 142.250.192.4, Dst: 172.30.1.37
- Transmission Control Protocol, Src Port: 80, Dst Port: 52280, Seq: 1, Ack: 169,
- Hypertext Transfer Protocol
 - HTTP/1.1 200 OK\r\n
 - [Expert Info (Chat/Sequence): HTTP/1.1 200 OK\r\n]
 - [HTTP/1.1 200 OK\r\n]
 - [Severity level: Chat]
 - [Group: Sequence]
 - Response Version: HTTP/1.1
 - Status Code: 200
 - [Status Code Description: OK]
 - Response Phrase: OK
 - Content-Type: text/html; charset=ISO-8859-1\r\n
 - [truncated]Content-Security-Policy-Report-Only: object-src 'none';base-uri
 - P3P: CP="This is not a P3P policy! See g.co/p3phelp for more info."r\n
 - Date: Mon, 06 Jan 2025 06:46:51 GMT\r\n
 - Server: gws\r\n
 - X-XSS-Protection: 0\r\n

The packet bytes pane shows the raw data in hexadecimal and ASCII.

Now can see the server column where all server name is showing.

Now we have to check how many Apache packets are there we can't count manually for each packet so we will apply another filter http.server contains "Apache"

Wireshark interface showing packet 113512 selected. The filter is 'http.response.code==200'. The packet details pane shows the following information:

- Frame 113512: 1111 bytes on wire (8888 bits), 1111 bytes captured (8888 bits) on
- Ethernet II, Src: AcrosserTech_46:43:a2 (00:02:b6:46:43:a2), Dst: LiteOnNetwor_6
- Internet Protocol Version 4, Src: 142.250.192.4, Dst: 172.30.1.37
- Transmission Control Protocol, Src Port: 80, Dst Port: 52280, Seq: 1, Ack: 169,
- Hypertext Transfer Protocol
 - HTTP/1.1 200 OK\r\n
 - [Expert Info (Chat/Sequence): HTTP/1.1 200 OK\r\n]
 - [HTTP/1.1 200 OK\r\n]
 - [Severity level: Chat]
 - [Group: Sequence]
 - Response Version: HTTP/1.1
 - Status Code: 200
 - [Status Code Description: OK]
 - Response Phrase: OK
 - Content-Type: text/html; charset=ISO-8859-1\r\n
 - [truncated]Content-Security-Policy-Report-Only: object-src 'none';base-uri
 - P3P: CP="This is not a P3P policy! See g.co/p3phelp for more info."r\n
 - Date: Mon, 06 Jan 2025 06:46:51 GMT\r\n
 - Server: gws\r\n
 - X-XSS-Protection: 0\r\n

The packet bytes pane shows the raw data in hexadecimal and ASCII.

Wireshark interface showing packet 113512 selected. The filter is 'http.server contains "Apache"'. The packet details pane shows the following information:

- Frame 113512: 1111 bytes on wire (8888 bits), 1111 bytes captured (8888 bits) on
- Ethernet II, Src: AcrosserTech_46:43:a2 (00:02:b6:46:43:a2), Dst: LiteOnNetwor_6
- Internet Protocol Version 4, Src: 142.250.192.4, Dst: 172.30.1.37
- Transmission Control Protocol, Src Port: 80, Dst Port: 52280, Seq: 1, Ack: 169,
- Hypertext Transfer Protocol
 - HTTP/1.1 200 OK\r\n
 - [Expert Info (Chat/Sequence): HTTP/1.1 200 OK\r\n]
 - [HTTP/1.1 200 OK\r\n]
 - [Severity level: Chat]
 - [Group: Sequence]
 - Response Version: HTTP/1.1
 - Status Code: 200
 - [Status Code Description: OK]
 - Response Phrase: OK
 - Content-Type: text/html; charset=ISO-8859-1\r\n
 - [truncated]Content-Security-Policy-Report-Only: object-src 'none';base-uri
 - P3P: CP="This is not a P3P policy! See g.co/p3phelp for more info."r\n
 - Date: Mon, 06 Jan 2025 06:46:51 GMT\r\n
 - Server: gws\r\n
 - X-XSS-Protection: 0\r\n

The packet bytes pane shows the raw data in hexadecimal and ASCII.

After applying filter go to Statistics > Endpoints

The screenshot shows the Wireshark interface with the filter `http.response.code==200` applied. The Statistics pane is open, and the 'Endpoints' option is selected under the 'Statistics' menu. The main pane displays a list of connections with columns: No., Time, Source, Host, Server, and Info. The connections are all from 182.156.100.0 to 182.156.100.255, all using HTTP/1.1 200 OK status, and all served by nginx.

No.	Time	Source	Host	Server	Info
184188	1035.532882	182.156.100.0	182.156.100.0	nginx	HTTP/1.1 200 OK
184271	1036.248374	182.156.100.1	182.156.100.1	nginx	HTTP/1.1 200 OK
184346	1036.837472	182.156.100.2	182.156.100.2	nginx	HTTP/1.1 200 OK
184437	1037.518468	182.156.100.3	182.156.100.3	nginx	HTTP/1.1 200 OK
184528	1038.127932	182.156.100.4	182.156.100.4	nginx	HTTP/1.1 200 OK
184606	1038.789738	182.156.100.5	182.156.100.5	nginx	HTTP/1.1 200 OK
184806	1039.515363	182.156.100.6	182.156.100.6	nginx	HTTP/1.1 200 OK
184938	1040.234151	182.156.100.7	182.156.100.7	nginx	HTTP/1.1 200 OK
185047	1040.951627	182.156.100.8	182.156.100.8	nginx	HTTP/1.1 200 OK
185183	1041.679724	182.156.100.9	182.156.100.9	nginx	HTTP/1.1 200 OK
185284	1042.302467	182.156.100.10	182.156.100.10	nginx	HTTP/1.1 200 OK
185375	1042.921916	182.156.100.11	182.156.100.11	nginx	HTTP/1.1 200 OK
185499	1043.572164	182.156.100.12	182.156.100.12	nginx	HTTP/1.1 200 OK
185602	1044.226030	182.156.100.13	182.156.100.13	nginx	HTTP/1.1 200 OK
185744	1044.934126	182.156.100.14	182.156.100.14	nginx	HTTP/1.1 200 OK
185863	1045.751288	182.156.100.15	182.156.100.15	nginx	HTTP/1.1 200 OK
185979	1046.671948	182.156.100.16	182.156.100.16	nginx	HTTP/1.1 200 OK
186094	1047.481762	182.156.100.17	182.156.100.17	nginx	HTTP/1.1 200 OK

It will show all connections

The screenshot shows the 'Endpoints - Ethernet' window in Wireshark. The 'Endpoint Settings' pane on the left has 'Limit to display filter' checked. The main pane displays a table of connections with columns: Address, Packets, Bytes, Total Packets, Percent Filtered, Tx Packets, Tx Bytes, Rx Packets, Rx Bytes, Country, and City. The connections are all from 182.156.100.0 to 182.156.100.255, all using HTTP/1.1 200 OK status, and all served by nginx.

Address	Packets	Bytes	Total Packets	Percent Filtered	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City
23.54.82.195	1	1 kB	12	8.33%	1	1 kB	0	0 bytes		
23.57.214.214	1	1 kB	12	8.33%	1	1 kB	0	0 bytes		
142.250.192.4	1	1 kB	63	1.59%	1	1 kB	0	0 bytes		
172.30.1.37	246	200 kB	35,660	0.69%	0	0 bytes	246	200 kB		
182.156.239.17	243	196 kB	1,719	14.14%	243	196 kB	0	0 bytes		

CONCLUSION: We have successfully analyzed the packets provided and solved the questions using wireshark.